H_2O_2 has a non-planar structure. The molecular dimensions in the gas phase and solid phase differ as shown in figure 4.5. Structurally, H_2O_2 is represented by the dihydroxyl formula in which the two OH^- groups do not lie in the same plane. One way of explaining the shape of hydrogen peroxide is that the hydrogen atoms would lie on the pages of a partly opened book, and the oxygen atoms along the spine. In the solid phase of molecule, the dihedral angle reduces to 90.2° due to hydrogen bonding and the O-O-H angle expands from 94.8 ° to 101.9°.

4.8 Hydrides

Hydrogen forms binary hydrides with many electropositive elements including metals and non-metals. It also forms ternary hydrides with two metals. E.g., $LiBH_4$ and $LiAlH_4$. The hydrides are classified as ionic, covalent and metallic hydrides according to the nature of bonding. Hydrides formed with elements having lower electronegativity than hydrogen are often ionic, whereas with elements having higher electronegativity than hydrogen form covalent hydrides.

Ionic (Saline) hydrides: These are hydrides composed of an electropositive metal, generally, an alkali or alkaline-earth metal, except beryllium and magnesium, formed by transfer of electrons from metal to hydrogen atoms. They can be prepared by the reaction of elements at about 400° C. These are salt-like, high-melting, white crystalline solids having hydride ions (H^-) and metal cations (M^{n+}).

$2\ Li + H_2 \rightarrow 2\ LiH$

$2\ Ca + 2H_2 \rightarrow 2\ CaH_2$

Covalent (Molecular) hydrides: They are compounds in which hydrogen is attached to another element by sharing of electrons. The most common examples of covalent hydrides of non-metals are methane, ammonia, water and hydrogen chloride. Covalent hydrides are further divided into three categories, viz., electron precise (CH_4, C_2H_6, SiH_4, GeH_4), electron-deficient (B_2H_6) and electron-rich hydrides (NH_3, H_2O). Since most of the covalent hydrides consist of discrete, small molecules that have relatively weak intermolecular forces, they are generally gases or volatile liquids.

Metallic (Interstitial) hydrides: Metallic hydrides are usually obtained by hydrogenation of metals and alloys in which hydrogen occupies the interstitial sites (voids). Hence, they are called interstitial hydrides; the hydrides show properties similar to parent metals and hence they are also known as metallic hydrides. Most of the hydrides are non-stoichiometric with variable composition ($TiH_{1.5\text{-}1.8}$ and $PdH_{0.6\text{-}0.8}$), some are relatively light, inexpensive and thermally unstable which make them useful for hydrogen storage applications. Electropositive metals and some other metals form hydrides with the stoichiometry MH or sometimes MH_2 (M = Ti, Zr, Hf, V, Zn).

4.9 Hydrogen Bonding

Hydrogen bonding is one of the most important natural phenomena occurring in chemical and biological sciences. These interactions play a major role in the

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